

The only solution to all the issues arising from cloud-based implementations is a cloud management platform (CMP). This provides a rich set of capabilities for discovery, template-based provisioning, orchestration and automation. It also enables operational monitoring and management, governance and cost optimisation across multiple public and private clouds as well as virtual and bare-metal servers.

A CMP is a comprehensive suite with integrated software tools that an enterprise can use to monitor and control cloud computing resources. An enterprise can use a CMP for either a private or public cloud, but CMPs facilitate toolsets for hybrid and multi-cloud models to centralise the control of various cloud-based infrastructures.

A wide range of tools is available to deal with these challenges and help companies efficiently operate applications and services in the cloud. Vendors offer a variety of cloud management tools that enable IT organisations to build, purchase, manage, monitor, track and optimise their cloud resources. With the help of these tools, organisations can save time and effort while allowing IT staff to focus on more strategic goals. These tools also help in monitoring users' interactions with the cloud infrastructure and in managing resource allocation.

In the cloud marketplace, different tools have unique feature sets. Capabilities include unified management across multiple clouds, integration with third-party tools for configuration and monitoring purposes, dashboards and reports for detailed information about resource consumption, notifications and alerts when predefined thresholds are reached, and controlled access to resources to avoid over usage or unauthorised access. By simplifying the management of cloud environments and reducing the complexity and cost of managing

multiple activities, the right cloud management tool can bring speed, flexibility, security and cost efficiency to any organisation.

So the important question is, "Which CMP is the best?" Well, there are both commercial and open source options. Nowadays, open source technology has become a central facet of cloud computing for many users around the world. In this article, we discuss the top open source cloud management platforms to assist admins, systems analysts, network security as well as cloud computing professionals select the best platform, based on the requirements of the enterprise.

Before we dive into the types of open source CMPs, let's discuss some of their broad capabilities and how they work.

According to analysis by Market Research Future (MRF), the global CMP market was valued at US\$ 8.18 billion in 2018 and is expected to reach US\$ 26.77 billion by 2022, registering a CAGR of 18.4 per cent during the forecast period. The market growth is attributed to the rising need for enterprises to have greater control over IT spending, the surge in the adoption of heterogeneous and multi-modal IT service delivery environments, the rapid deployment of virtualised workloads, and improved operational efficiency. On the other hand, insufficient technical expertise and the rising security concerns for platforms developed in-house are some of the factors expected to hinder the market growth during the assessment period.

How cloud management platforms work

A CMP is deployed into existing cloud environments as a virtual machine (VM) consisting of a database and server. The server communicates with application programming interfaces (APIs) to connect the

database and virtual resources held in the cloud. The database collects the information on how the virtual infrastructure is performing and sends an analysis to the Web interface, where systems administrators can analyse the cloud performance. The whole interconnectivity relies on the operating system, which commands all the different technologies that make up clouds and also deploys cloud management tools.

A CMP should be capable of the following things.

- **Strong integration with IT infrastructure:** CMPs should be customised as per the enterprise's needs, and must meet the requirements of the operating systems, apps, storage frameworks and anything else running in the cloud.
- **Automating manual tasks:** CMPs should have self-service capabilities to automate everything, with no human involvement.
- **Cost management:** CMPs should assist organisations with precision cost forecasting and reporting to easily use and manage all sorts of cloud services.
- **Service management:** They should assist the IT team to monitor cloud-based services to help in capacity planning, workload deployment, asset management and incident management.
- **Governance and security:** CMPs should enable administrators to enforce policy-based control of cloud resources, and offer security features like encryption as well as identity and access management.

Top open source CMPs

The following are the top open source cloud management platform providers.

Apache CloudStack

Apache CloudStack is an open source, multi-hypervisor, multi-tenant, high-availability Infrastructure-as-